Harnessing AI for Ecosystem Surveillance: A Case Study on Wetland Mapping with Transformer Models

Arden Lu
International Division
Shanghai High School
Shanghai, China
3024505357@qq.com

Qiwei Han*
Nova School of Business and Economics
Nova University Lisbon
Lisbon, Portugal
qiwei.han@novasbe.pt
*Corresponding author

2024 13th International Conference on Communications, Circuits and Systems (ICCCAS) | 979-8-3503-8627-1/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICCCAS62034.2024.10652836

***Abstract*—Wetlands play a crucial role in maintaining ecological balance, making their accurate classification and monitoring imperative. This paper presents a novel approach to wetland classification using satellite imagery, harnessing the power of state-of-the-art deep learning architectures. We introduce the application of the advanced Transformer and Convolutional Neural Network models, to the domain of wetland classification, demonstrating its potential in handling intricate spatial data. To further enhance the model's performance, we employ a transfer learning strategy, fine-tuning the transformer on the comprehensive BigEarthNet-S2 satellite image dataset. This approach capitalizes on the model's pre-trained knowledge while tailoring it to the specific characteristics of wetland imagery. Our research also introduces a modularized workflow, designed to be both robust for the current classification task and adaptable for future advancements in the field. The results showcase a significant improvement in classification accuracy. This research not only sets a new standard in the satellite-based monitoring of wetlands but also contributes to the broader dialogue on resource and energy conservation.**

Keywords—wetland recognition, deep learning, transformer, transfer learning

I. Introduction

Wetlands, defined by their saturation with water, either permanent or seasonal, are acknowledged as some of the most productive landscapes on Earth [4]. These biologically diverse ecosystems render a multitude of ecological, economic, and social benefits that are crucial for both human prosperity and environmental health. Wetlands provide vital habitats for a multitude of species, including those that are rare or endangered, and are crucial breeding grounds for fish, amphibians, and migratory birds, enriching biodiversity [1]. Additionally, their role in water purification through phytoremediation processes underscores their importance in sustaining water quality and aquatic ecosystems [2]. As natural disaster mitigators, wetlands absorb floodwaters, mitigate soil erosion, and reduce property damage, with coastal wetlands like mangroves being especially significant against storm surges and hurricanes [1] . In terms of climate change, wetlands are substantial carbon sinks, thus playing an integral role in carbon cycling [5]. Economically, wetlands are beneficial for tourism, fisheries, and agriculture, supporting local communities by offering employment and bolstering food security [6].

Recognizing these benefits underscores the urgency for wetland conservation, aligning with the United Nations' Sustainable Development Goals pertaining to climate action, life on land, and aquatic life [7]. Conservation and restoration efforts are imperative for the continued delivery of essential ecosystem services, biodiversity protection, and global sustainability advancement. These efforts must be undergirded by comprehensive policy-making, research, education, and public engagement to ensure enduring preservation and planetary welfare. However, the relentless expansion of industry and urbanization has precipitated the decline or disappearance of many wetland areas, prompting an urgent call for their monitoring and protection [8].

The accurate delineation of wetlands is paramount in conservation strategies. Traditional detection has relied upon ecologist-led field surveys, which, while precise, are labor-intensive and thus inefficient [9]. Conversely, the advent of satellite remote sensing and image processing technologies has enhanced the efficiency of wetland analysis, with increasing reliance on these methods since their inception in the 1980s. The evolution of these technologies has seen a shift from classic image processing to the incorporation of machine learning algorithms, enhancing classification accuracy through methods such as random forests and support vector machines (SVM). In addition, convolutional neural networks (CNN) with transfer learning are being explored for their potential in classifying various wetland types [4].

Prompted by these developments, this paper advocates the application of deep learning techniques for the processing of satellite imagery to detect wetlands with high accuracy. The major contributions are as follows: We investigate both CNN and transformer-based structures, incorporating transfer learning to expedite training. Our proposed methods aim to not only outperform existing techniques in both recognition precision and computational efficiency, but also raises significant implications on the efficient use of renewable natural resources and the mitigation of environmental impacts.

The subsequent sections of this paper are arranged as follows: Section II summarizes the background and synthesizes related literature; Section III details the datasets for training and testing neural networks; Section IV outlines the methodologies; Section V discusses experimental results; and Section VI concludes with reflections and prospects for subsequent inquiries..

II. Related Works

The application of satellite remote sensing for wetland surveying traces its roots back to the 1980s. During this period, the primary focus was on detecting permanently flooded or exposed water ponds, which were relatively easier to identify

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[10]. The 1990s saw a shift in focus towards coastal tidal marshes, which became the most frequently studied wetlands during that decade [11].

The advent of the 21st century marked significant advancements in computer technologies and image processing techniques, which have greatly improved the capabilities of satellite image processing for wetland detection. Matthews et al. employed passive microwave signals and a vegetation cover model to calculate the coverage of seasonal flooding, achieving a high degree of accuracy [12]. Frohn et al. utilized image segmentation and object-oriented classification methods on Landsat-7 data to detect isolated wetlands in the U.S., reporting an accuracy range of 88% to 97% [13]. Lane et al. used Worldview-2, an eight-band satellite system, to identify various classes of wetland habitats in Lake Baikal, Russia [14]. Their work demonstrated that the use of less common spectral bands could enhance the differentiation between different types of wetlands, achieving an accuracy of 86.5% for 22 classes. Hu et al. conducted a high spatial resolution satellite imagery analysis on the Hangzhou Xixi Wetland and found a significant decrease in wetland areas from 2000 to 2013 [15].

The success of machine learning techniques and in particular deep learning in various domains [16], such as computer vision, natural language processing, and speech recognition, has piqued interest in their application for wetland detection. Li et al. compared the accuracies of different machine learning algorithms for land-cover classification, demonstrating the superiority of machine learning techniques over traditional methods [17]. Specifically, they found that random forest and expert-based approaches achieved accuracies of 83.9% and 90.3%, respectively, outperforming the maximum likelihood method, which had an accuracy of 63.3%. Murray et al. applied machine learning classifiers to analyze changes in tidal flats, tidal marshes, and mangroves [18]. They reported a net change of -4000 km² over two decades, attributing 27% of this change to direct human activities. Thamaga et al. proposed using machine learning algorithms to detect changes in wetland vegetation in Sub-Saharan Africa [19]. Liu et al. employed DeepLabV3+ and HRNet algorithms, along with transfer learning, to classify marsh vegetation [20]. Their findings suggested that image combination improves classification accuracy for low-resolution images but decreases it for higher-resolution images. Munizaga et al. evaluated the feasibility of support vector machine and random forest algorithms in different spectral bands, achieving accuracies of 88% and 90%, respectively [20]. Jafarzadeh et al. provided a comprehensive review of machine learning and remote sensing techniques used over the decades, offering insights into suitable models and data types for machine learning in wetland detection [21].

III. Data Preprocessing and Augmentation

A. Data Sources and Collection

The primary data source for this study is BigEarthNet, a comprehensive dataset collected by the Remote Sensing Image Analysis (RSiM) Group and the Database Systems and Information Management (DIMA) Group at Technische Universität Berlin (TU Berlin) [22]. It contains satellite imagery from Sentinel-1 (BigEarthNet-S1) and Sentinel-2 (BigEarthNet-S2), collected between June 2017 and May 2018 over 10 European countries respectively. Each dataset in BigEarthNet contains 590,326 image patches. These patches are classified into Level-3 CLC (CORINE Land Cover) classes, including but not limited to inland wetlands and marine wetlands, which are the focus of this study. The image patches come in various resolutions: 1) 120x120 pixels for 10-meter bands; 2) 60x60 pixels for 20-meter bands and 3) 20x20 pixels for 60-meter bands.

We choose BigEarthNet-S2 dataset for this study given its fully labeled images on land cover classes. In particular, Sentinel-2 is a high-resolution, multispectral imaging mission from the European Union's Copernicus Program. It consists of two identical satellites, Sentinel-2A and Sentinel-2B, each equipped with a Multi-Spectral Instrument (MSI) that samples 13 spectral bands at various spatial resolutions. These bands range from visible to near-infrared and short-wave infrared spectrums. We retrieved 3 bands red (B4), green (B3) and blue (B2) to represent RGB images so that it can fit with general machine learning models.

BigEarthNet employs the CORINE Land Cover database (CLC 2018) for labeling. Initially, a 43-class nomenclature was used, but it has been updated to a more streamlined 19-class nomenclature designed for machine learning applications. This change was necessary due to the limitations of some CLC classes that could not be easily identified using Sentinel-2 images alone. Among 19-class nomenclatures, two types of wetlands exist: inland wetlands and coastal wetlands and our goal is to classify patches of satellite imagery into accurate land cover classes, such that later it can be used to detect the wetland classes accurately.

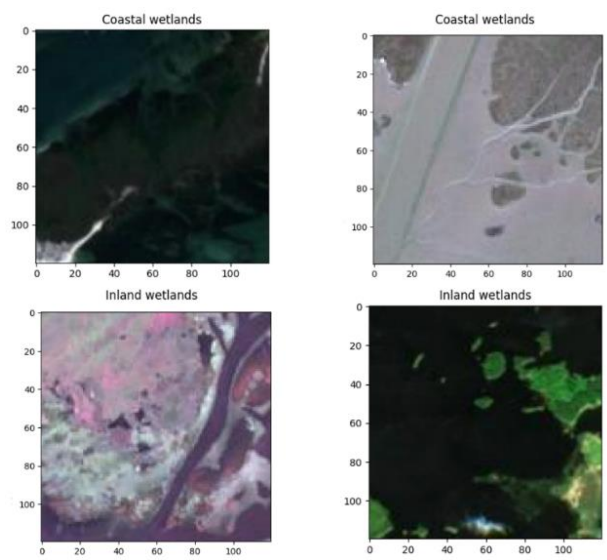

Fig. 1. Examples of satellite images categorized as coastal wetlands and inland wetlands.

B. Image Preprocessing

Before utilizing the image patches for model training and evaluation, a series of preprocessing steps are essential to ensure optimal performance. The preprocessing pipeline consists of two main components: image transformations and data augmentation.

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1) Image transformations

- Resizing: The images are resized to a uniform dimension, facilitating batch processing during model training.
- Normalization: The color channels (Red, Green, Blue) are normalized using pre-calculated mean and standard deviation values. This step is crucial for ensuring consistent input data and improving model convergence.

2) Data augmentation

To enhance the model's robustness and potentially improve its generalization capabilities, we apply data augmentation techniques during the training phase. These techniques include:

- RandomHorizontalFlip: Randomly flips the image horizontally.
- RandomResizedCrop: Randomly crops and resizes the image.
- CenterCrop: Crops the center of the image.

To streamline the preprocessing workflow and ensure compatibility with the chosen model architecture, we built an automatic image processor that serves as a minimal yet effective tool for preparing images for both training and inference later. Furthermore, to efficiently manage computational resources, we apply these preprocessing steps on-the-fly, meaning they are executed as the images are loaded into RAM. This would dynamically applies the aforementioned transformations and augmentations to the whole image dataset..

IV. METHODS

Wetland detection is fundamentally a classification task, which can be addressed using various deep learning architectures. In this study, we focus on Convolutional Neural Networks (CNNs) and Transformer-based models, specifically examining VGG, ResNet, Swin Transformer, and Vision Transformer (ViT). This section outlines the architecture of these neural networks and describes the training methodology employed.

A. Neural Network Structures

1) VGG

VGG, or VGGNet, is a deep CNN architecture known for its simplicity and effectiveness [23]. Two popular variants of VGG are VGG-16 and VGG-19, where the numbers indicate the total layers in the network. The architecture comprises the following key components:

- Convolution layer: VGG employs a 3x3 receptive field, the smallest size capable of capturing the spatial hierarchies in an image. Additionally, a 1x1 convolution filter is used for linear transformations. The stride is set to 1 to preserve spatial dimensions post-convolution.
- The Rectified Linear Unit (ReLU) serves as the activation function, significantly reducing training time compared to its predecessors like AlexNet.
- Fully connected layer: VGGNet incorporates three fully connected layers. The first two layers contain 4,096 channels each, while the last layer has 1,000 channels, corresponding to each class.

Fig. 2. VGG-16 architecture adapted from [23].

2) ResNet (Residual Networks)

ResNet introduces the concept of residual learning to mitigate the vanishing gradient problem in deep networks [24]. It uses "skip connections" that bypass one or more layers to make training deep networks more efficient.

- Skip Connections: These are the shortcuts that allow the gradient to be directly backpropagated to earlier layers, mitigating the vanishing gradient problem commonly observed in deep networks.
- Bottleneck Architecture: To make the network more computationally efficient, ResNet often uses a bottleneck design where three layers are stacked together: 1x1, 3x3, and 1x1 convolutions. The 1x1 layers are responsible for reducing and then increasing (restoring) dimensions, leaving the 3x3 layer a bottleneck with smaller input/output dimensions.
- Global Average Pooling: Instead of using fully connected layers, ResNet employs global average pooling to reduce the spatial dimensions of the 3D tensor to generate class scores, making the network less prone to overfitting.
- Batch Normalization: ResNet uses batch normalization to stabilize and speed up the training of deep networks. This is applied after every convolutional layer.

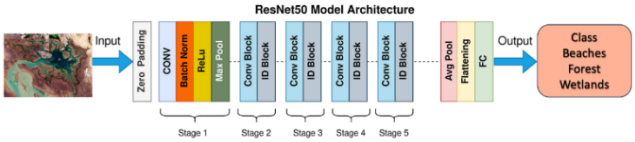

Fig. 3. ResNet architecture adapted from [24].

3) Vision Transformer (ViT)

Vision Transformer (ViT) adapts the Transformer architecture [25], originally designed for natural language processing, for computer vision tasks [26]. It divides an image into a sequence of patches and processes them in a manner similar to how words are processed in text.

- Patch Embedding: The input image is divided into fixed-size patches, which are then linearly embedded into a 1D sequence. These serve as the input tokens for the Transformer model.
- Positional Encoding: Since the Transformer architecture doesn't have a built-in sense of order or position, positional encodings are added to the patch embeddings

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to give the model information about the location of each patch in the image.

- Multi-Head Self-Attention: ViT uses the multi-head self-attention mechanism to weigh the importance of different parts of the input image, allowing the model to focus on more relevant areas.
- Layer Normalization: Similar to batch normalization in CNNs, layer normalization is used in each Transformer layer to stabilize the activations.
- Hybrid Models: ViT can be combined with other architectures like CNNs for the feature extraction phase, creating a hybrid model that leverages the strengths of both architectures.

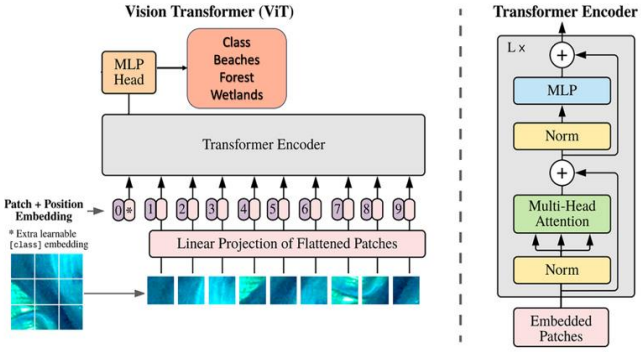

Fig. 4. Vision Transformer (ViT) architecture adapted from [26].

4) Swin Transformer and Swin Transformer v2

Swin Transformer is a novel deep learning architecture, which is based on self-attention mechanism and has exhibited extraordinary performance in computer vision applications [27]. Compared to classical CNN, Swin Transformer introduced an innovative way to process the image. The new features of Swin Transformer are summarized as follows:

- Windowed Self-Attention: Unlike traditional CNNs, which are limited to capturing local spatial hierarchies, Swin Transformer uses windowed self-attention to focus on distant positions in the image, thereby capturing long-range dependencies.
- Hierarchical Architecture: The image is divided into smaller blocks or "mini-images." Self-attention is applied individually to these blocks, reducing computational complexity while still capturing global context through shifted windows and hierarchical structures.

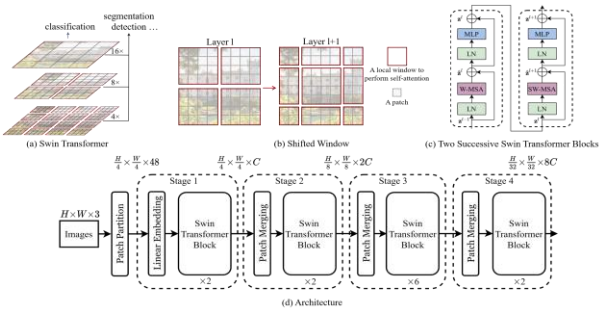

Fig. 5. Swin transformer architecture adapted from [27].

Moreover, Swin Transformer v2 [28] made several adaptations to original Swin Transformr architecture as follows:

- Scaled Cosine-Attention: Swin Transformer v2 replaces the original dot-product attention with the scaled cosine attention mechanism, which leads to higher scalability as the attention values are less likely to fall into extremes.
- Log-spaced continuous relative position bias: Swin Transformer v2 effectively transfers pre-trained models using low-resolution images to downstream tasks with high-resolution inputs.
- Self-supervised pre-training method: SimMIM is adopted to reduce the need for large amount of labeled data [34].

5) ConvNeXT and ConvNeXT v2

Although ViT and other transformer architecture showed superior performance over CNN architecture, Liu et al. in [29] argued that convolution operations are still useful in generic image classification tasks and proposed a family of models constructed entirely from CNN moduels, named ConvNeXT and achieved comparable performance in image classifications. In particular, ConvNeXT used a multi-stage design based on ResNet and introduced a few tricks such as larger kernel size and fewer activation functions and normalization layers in training process as follows:

- Adjusted number of blocks: the number of blocks in ResNet has been adjusted to align with transformer architecture with the same stage compute ratio.
- “Patchify” the stem cell: similar to transformer architecture, the input images are downsampled to an appropriate feature map size, known as the patchify strategy, using convolutional layers.
- ResNeXT-ify: ResNeXT method is adopted [30] to introduce the grouped depthwise convolution where the number of groups equals the number of channels. This method is similar to self-attention used in Transformer architecture and leads to improved performance.

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B. Transfer Learning Approach

In addition to the architecture of the neural networks, another crucial aspect of our experimental setup was the adoption of a transfer learning approach. Transfer learning allows us to leverage pre-trained models to accelerate the training process and potentially achieve higher performance. This is particularly useful when the dataset is not large enough to train a deep neural network from scratch effectively.

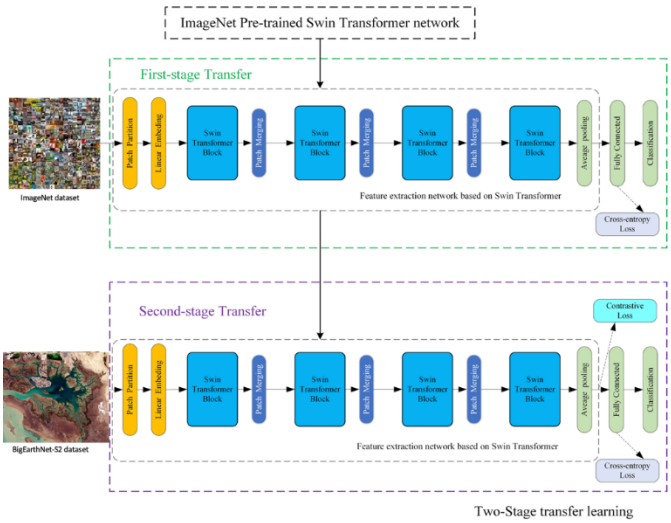

Fig. 6. Two-stage transfer learning framework using Swin Transformer as both backbone and fine-tuned model.

1) Pre-trained Models

We utilized pre-trained models available in the Keras library for the VGG and ResNet architectures, and specific pre-trained versions for the Vision Transformer (ViT), Swin Transformer and ConvNeXT models. The following pre-trained models were used:

- VGG16: keras.applications.vgg16.VGG16
- VGG19: keras.applications.vgg19.VGG19
- ResNet-50: keras.applications.resnet50.ResNet50
- Vision Transformer (ViT): vit-base-patch16-224-in21k
- Swin Transformer: swin-large-patch4-window7-224
- Swin Transformer v2: microsoft/swinv2-base-patch4-window16-256
- ConvNeXT: facebook/convnext-base-224-22k-1k
- ConvNeXT v2: facebook/convnextv2-base-1k-224

2) Fine-Tuning with BigEarthNet-S2

Each pre-trained model was fine-tuned using the BigEarthNet-S2 dataset. Fine-tuning involves taking a pre-trained model and continuing the training process on a new dataset, allowing the model to adapt to the specific characteristics of the new data. In our case, the BigEarthNet-S2 dataset was used for this purpose, which contains Sentinel-2 images specifically related to wetland classification.

3) Fine-tuning procedure

The fine-tuning was performed in two steps:

- Feature Extraction: Initially, the layers of the pre-trained models were frozen, and only the final fully connected layer was trained on the BigEarthNet-S2 dataset. This allows the model to learn the high-level features specific to wetland classification.
- Fine-Tuning All Layers: After the feature extraction phase, all layers were unfrozen, and the entire network was trained for a few more epochs. This allows the model to adapt its pre-learned features to the new dataset better.

V. Experimental Results

The experiments were conducted on a high-performance computing cluster equipped with 1 NVIDIA Tesla V100 GPU. This GPU is known for its high computational capabilities, making it ideal for deep learning tasks. The software stack included and PyTorch 2.0 and Torchvision for implementing and training the neural network models.

The primary metric used for evaluating the performance of the models was classification accuracy. Accuracy is defined as the ratio of correctly classified instances to the total number of instances in the dataset. It is a widely-used metric in image classification tasks and provides a straightforward way to compare different models.

After fine-tuning, the models demonstrated a noticeable improvement in classification accuracy, further validating the effectiveness of the transfer learning approach for this specific task. The classification accuracies obtained from the different models are as follows:

TABLE I. Performance Comparisons in Terms of Model Complexity (#Parameters) and Accuracy

Model name	# parameters	Model performance (accuracy)
VGG-16	~138 million	63.28%
VGG-19	~140 million	63.56%
Resnet-50	~23 million	67.23%
ViT	86 million	72.28%
Swin Transformer	3 billion	72.60%
Swin Transformer v2	3 billion	72.90%
ConvNext	659 million	72.30%
ConvNext v2	659 million	73.10%

- VGG16: Achieved an accuracy of 63.28%. While VGG16 is a robust model, its performance was relatively lower compared to the more modern architectures. This could be due to its deeper structure, which might make it more susceptible to overfitting for this particular task.
- VGG19: Slightly outperformed VGG16 with an accuracy of 63.56%. The additional layers in VGG19 may have captured more complex features, but the improvement was marginal.
- ResNet-50: Demonstrated a significant improvement over the VGG models, achieving an accuracy of 67.23%. The use of residual connections likely helped the model

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learn more complex representations of the data, leading to better performance.

- Vision Transformer (ViT): Achieved an accuracy of 72.28%, outperforming all the previous models. This indicates that the self-attention mechanisms and the ability to capture long-range dependencies in the image are beneficial for this task.
- Swin Transformer: Recorded the higher accuracy of 72.60%, slightly outperforming ViT. The window-based self-attention and hierarchical structure likely contributed to this superior performance.
- ConvNext: Achieved the highest accuracy of 73.10%, outperforming both ViT and Swin Transformer with less model complexity and training time. This model is more appealing given its better performance and higher scalability when training with large-scale satellite image dataset.

The results indicate that Transformer-based models, specifically ViT and Swin Transformer, outperform traditional CNN-based models like VGG and ResNet for this particular wetland classification task. This could be attributed to the Transformers' ability to capture long-range dependencies and more complex features in the images. However, it's worth noting that the difference in performance between ViT and Swin Transformer was minimal, suggesting that both models are highly effective for this application.

VI. Discussion and Conclusions

Wetlands are vital ecosystems that provide numerous ecological services, from biodiversity conservation to water purification and carbon sequestration. However, they are increasingly under threat due to human activities and climate change. Accurate and timely monitoring of these ecosystems is essential for their preservation, and satellite remote sensing offers a scalable solution. By comparing various neural network architectures—VGG, ResNet, Vision Transformer (ViT), Swin Transformer, and the newly introduced ConvNext—on the BigEarthNet-S2 dataset, this research aimed to identify the most effective models for classifying wetlands through satellite imagery.

Our findings indicate a clear advantage of transformer-based models over traditional convolutional neural networks (CNNs) for this specific task. Both ViT and Swin Transformer outperformed VGG and ResNet models, achieving classification accuracies of 72.28% and 72.60%, respectively. The results suggest that the self-attention mechanisms in transformers are particularly effective in capturing the complex features and long-range dependencies inherent in wetland imagery. However, the introduction of ConvNext, and its subsequent version, ConvNext v2, marks a significant milestone, achieving the highest accuracies of 72.30% and 73.10%, respectively. These results underscore the evolving nature of deep learning architectures and suggest that advancements in CNN models, such as ConvNext, can rival or even surpass transformer models in specific applications like wetland classification. The study also demonstrated the efficacy of a transfer learning approach, utilizing pre-trained models and fine-tuning them on the BigEarthNet-S2 dataset, thereby accelerating the training process and enhancing model performance.

The significance of these findings lies in their potential to revolutionize the way we approach wetland monitoring. The high accuracies achieved by the transformer-based models could translate into more precise and timely interventions for wetland preservation. For instance, accurate classification can help in identifying areas that are degrading at a faster rate, enabling conservationists and policymakers to take targeted actions. Moreover, the use of satellite imagery makes this approach scalable and cost-effective, allowing for frequent monitoring of wetlands across large geographical areas. The paper also contributes to the growing body of literature on transfer learning in environmental science. By fine-tuning pre-trained models on the BigEarthNet-S2 dataset, we demonstrated that transfer learning could significantly improve classification performance.

In this paper, we have made several contributions to the domain of wetland classification using satellite imagery. Firstly, we have embraced the state-of-the-art transformer architecture, specifically the Swin Transformer, which has been predominantly recognized for its prowess in natural language processing, and adeptly applied it to the realm of wetland classification. This innovative application showcases the versatility and potential of transformer models in handling complex spatial data. Secondly, our work leverages the power of transfer learning, fine-tuning our model on the expansive and labeled satellite image dataset, BigEarthNet-S2. This approach not only harnesses the pre-trained knowledge of the model but also tailors it to the specific nuances of wetland imagery, resulting in a significant enhancement in model performance. Lastly, we have designed a modularized workflow that stands as a robust framework for this classification task. Its modular nature ensures that it remains adaptable and extensible, ready to incorporate future advancements and developments in the field. Through these contributions, our paper sets a new benchmark and provides a comprehensive blueprint for future endeavors in satellite-based wetland classification.

However, the study is not without its limitations. One of the primary constraints is the focus on European wetlands through the BigEarthNet-S2 dataset. The generalizability of the models to wetlands in other geographical and climatic conditions remains an open question. Additionally, the study did not explore the impact of different preprocessing techniques or data augmentation strategies, which could potentially improve model performance further.

Looking ahead, future work could aim to validate the models on wetlands from diverse geographical regions and under varying environmental conditions. Incorporating temporal data could also provide insights into the seasonal changes in wetlands, offering a more dynamic understanding of these ecosystems. Another avenue for research could be the integration of additional data sources, such as topographical or hydrological data, to create more robust and comprehensive models for wetland classification.

In conclusion, this study highlights the significant potential of both transformer and advanced CNN architectures, like ConvNext, in the field of environmental science for wetland monitoring and classification. While transformers have shown

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promise, the exceptional performance of ConvNext models indicates that CNN architectures remain highly relevant and potent for image-based classification tasks. As machine learning technology continues to evolve, so too will our ability to monitor and preserve vital ecosystems like wetlands, leveraging these advancements for the betterment of environmental conservation efforts.

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